

# Assignment 2

**\*\*You have to submit handwritten answers and solutions and submit one pdf containing all the solutions.\*\***

**Question :**

**Consider the following Python function:**

```
def process_numbers(n):
    i = 1
    total = 0

    while i <= n:
        if i % 2 == 0:
            total += i          # Case A: even
        else:
            if i % 3 == 0:
                total -= i      # Case B: odd & divisible by 3
            else:
                total += 1      # Case C: odd & not divisible by 3

        # secondary branching affecting side effects only
        if i % 5 == 0 and total % 2 == 1:
            print("checkpoint") # observable side effect for testing

        i += 1

    if total > n:
        print("Larger")
    else:
        print("Smaller or Equal")
```

# Tasks

## 1. Construct the Control Flow Graph (CFG)

- Identify basic blocks and label all edges.
- Show loop back-edges and each branching point explicitly (including the nested `if` and the post-iteration `if`).

## 2. Compute Cyclomatic Complexity (M)

## 3. Identify a Basis Set of Independent Paths

- List at most M linearly independent paths that together cover all nodes/edges.
- Use clear path notation referencing your CFG node labels.

## 4. Design Test Cases

- Choose concrete `n` values (and explain why) so that executing the function covers each basis path.
- Ensure your tests also exercise the `checkpoint` print (the `i % 5 == 0` and `total % 2 == 1` branch) at least once.

## 5. Path Feasibility Discussion

- Briefly justify that each listed path is feasible (or explain constraints if not).